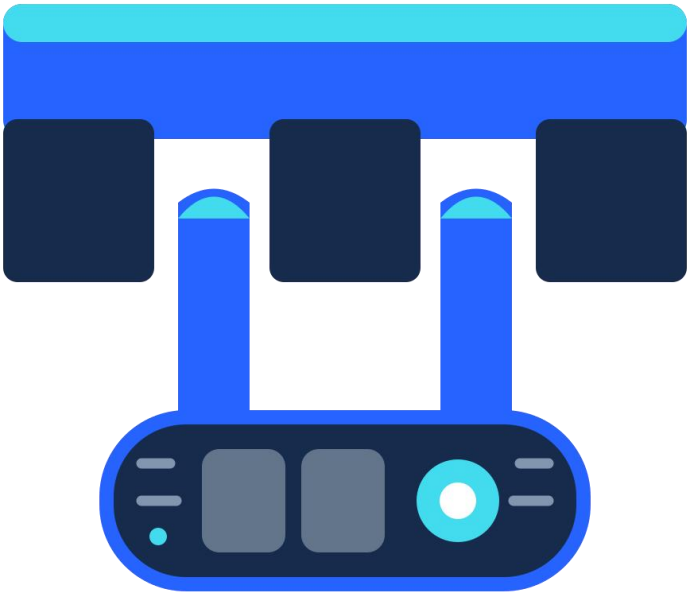
MRDVS



# AW3 Smart Docking

## Smart Docking User Manual

**Semantic and geometric positioning for autonomous forklift pallet docking.**

Zhejiang MRDVS Technology Co., Ltd.

V0.1 · 2026-09-28

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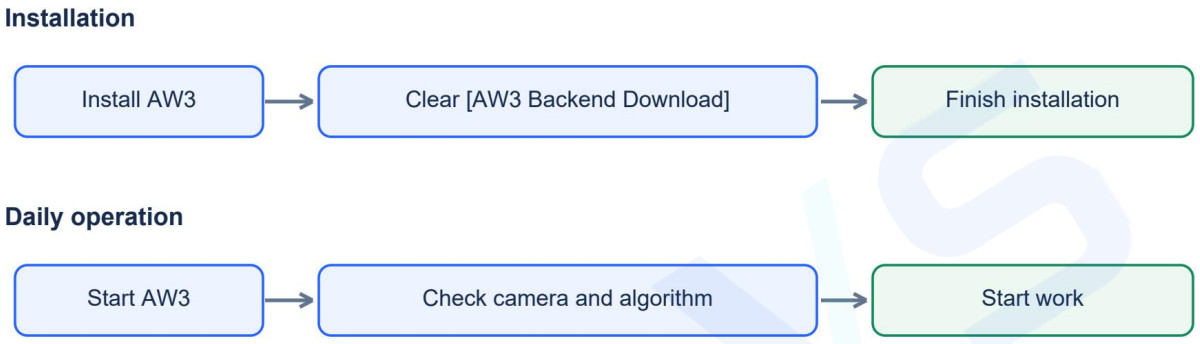
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# 1 Product Overview

Smart Docking runs on the AW3 algorithm platform and supports 2D semantic and pure 3D positioning. It outputs the pallet fork-entry reference position, orientation and positioning status for autonomous forklift docking.

## 1.1 Operating Prerequisites

1. Install AW3. No AW3 backend is required; clear the [AW3 Backend Download] checkbox.
2. Connect the camera and select the [Pallet Docking] task in Camera Groups.
3. Check that images and point clouds update continuously and the algorithm runs normally.



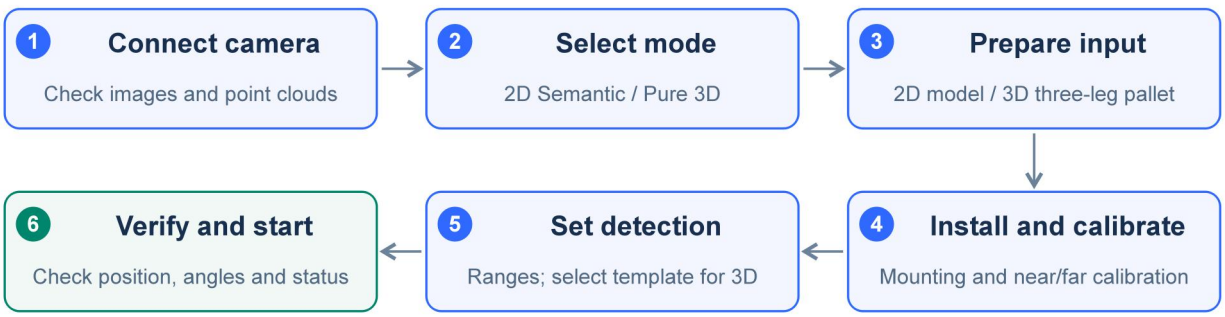

AW3 installation and daily startup: select and run Smart Docking in the platform

**Streaming errors** If AW3 streaming fails after configuring a camera in LxCameraViewer, open [Device Connection / Base Camera] > [Central Service] > [Restart current center]. No restart is needed if streaming is normal.

For AW3 installation, device connection and general communication setup, see the AW3 Deployment User Manual.

## 1.2 First-Use Workflow

Before deployment: collect site-pallet RGB images for 2D model training and validation. 3D requires no training-data preparation.



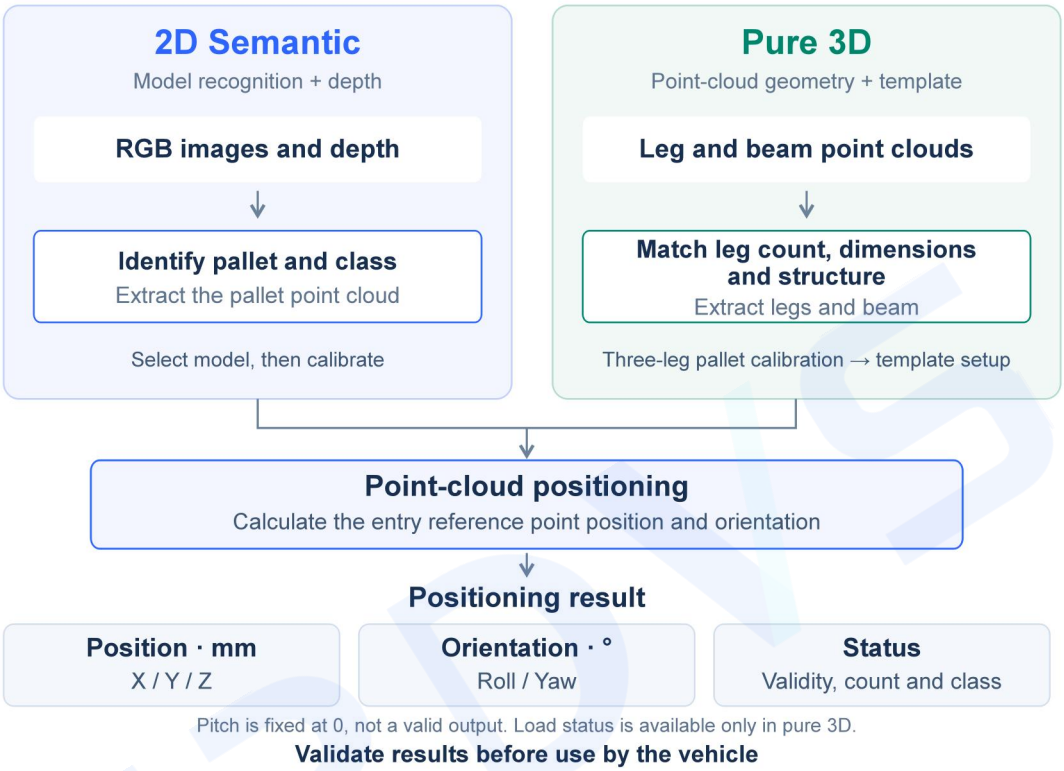

Before starting: secure the camera; keep the entry face and floor visible; confirm pallet dimensions and camera working distance suit the task.

Check the conditions in Section 2.2 before use. For operating steps, see Section 3.2 for 2D or Section 3.3 for 3D.

# 2 Before You Start

## 2.1 Choose a Positioning Mode

Both modes output pallet position, orientation and positioning status. 2D uses image-model recognition; pure 3D matches point-cloud geometry. Choose a mode using the capability comparison below.




2D semantic positioning and pure 3D positioning

The valid positioning output has five axes: X, Y, Z, Roll and Yaw. Position is in mm and angles are in °; Pitch is not a valid output. Load status applies only to pure 3D mode with load detection enabled. Validate results before vehicle use.

| Function | 2D Semantic Positioning | Pure 3D Positioning |
|---|---|---|
| 2-, 3- and 4-leg pallets | Supported with a suitable model | Supported with a geometry template |
| With / without lower beam | Classification supported | Not classified |
| Load detection | Not supported | Supported when enabled |

For initial setup, follow Section 3.2 for 2D or Section 3.3 for pure 3D. Complete only the workflow for your selected mode, then verify the results in Section 3.4.

## 2.2 Operating Conditions and Limits

### Check the camera, distance and cycle time

**Detection distance limit**




**Complete detection cycle**

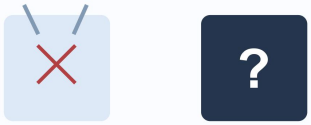
| 2D Semantic Positioning | | Pure 3D Positioning |
|---|---|---|
| S11 + RK3588 | M4 Mega | 200 ms max. |
| 210 ms max. | ~500 ms max. | S11 + RK3588 and M4 Mega |

Distance limits are the same for both modes. Validate the usable range with site pallets. The minimum recognition distance depends on pallet size and type.

Plan by mode and platform. Validate on site and allow margin; not for real-time motion feedback.

## Operating Conditions and Limits

### Pallet material




Reflective surfaces are unsuitable.
Assess black or low-reflectance pallets.

### Pallet structure and visibility



Show the full entry face, legs and beams.
Occlusion and missing parts affect positioning.
Thin-legged, deformed or self-built pallets require assessment before use.

### 2D model suitability




Train with RGB images of site pallets.
Coverage depends on training and validation.

### Entry-face angle




Within 15° relative to the camera.
Angles over 15° are filtered.

### 3D recognition height




Camera-to-floor height must match calibration.
A height change requires ground recalibration.

### 3D roll tilt

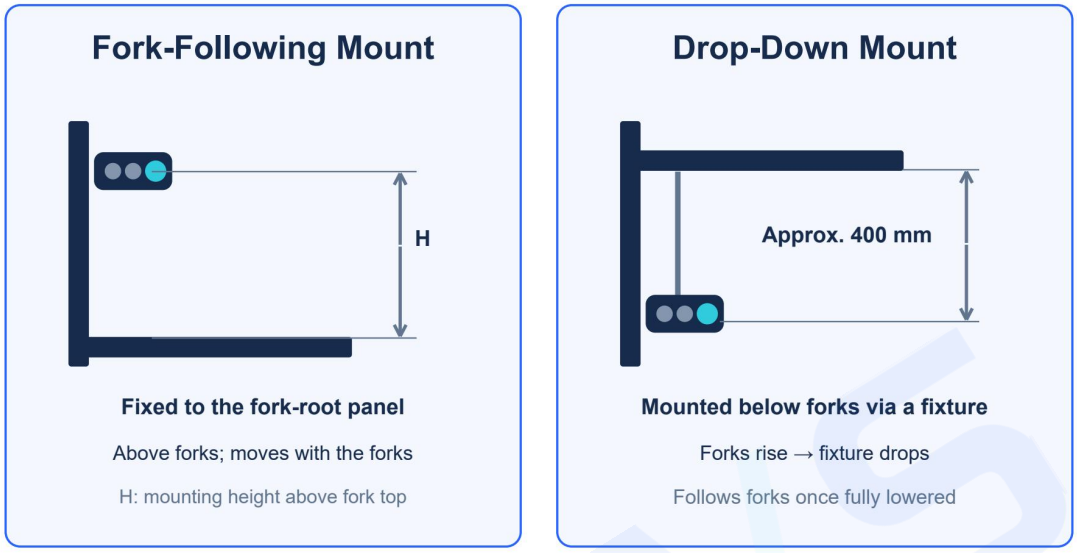


Roll tilt may prevent recognition.
Roll output does not mean all tilts are supported.

**Safety** Positioning results do not replace vehicle obstacle avoidance, emergency stops or safety interlocks.

## 2.3 Installation Requirements

Use follow-fork or drop-down mounting. Choose S11 + RK3588 for closer-range tasks, or M4 Mega with onboard compute for longer-range tasks. Check fixing, height and visibility against the vehicle structure.

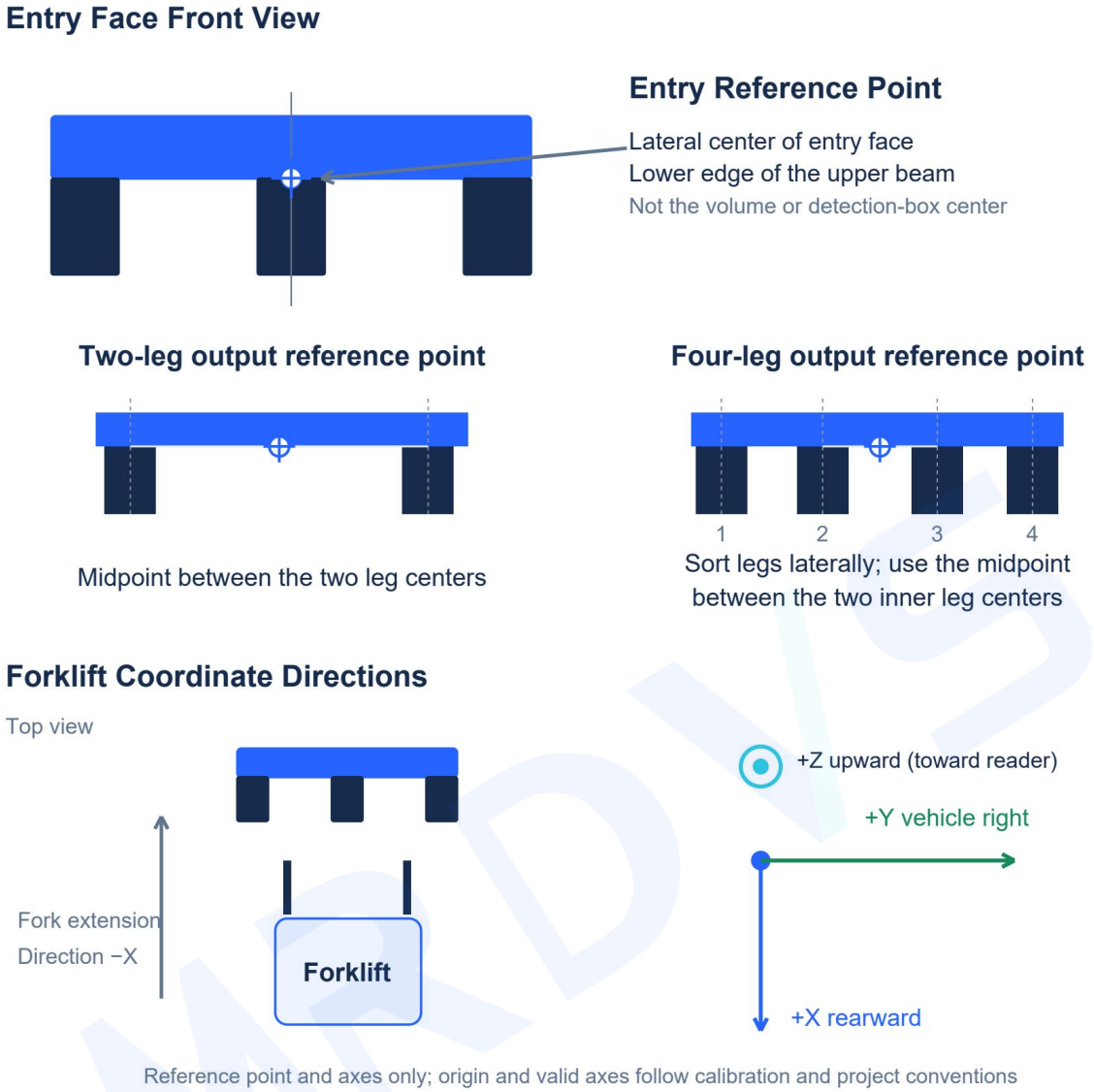



Fork-following and drop-down mounts (not to scale)

| Check | Requirement |
|---|---|
| Center and level | Mount near the centerline between the two forks, within 50 mm laterally. Keep the camera level, its front vertical to the floor and pitch near zero. |
| Fixation and power | Secure brackets and cables against pulling, pinching and impact during mast travel. Follow the hardware manual; check power, network, streaming, and external compute-unit fixation and cooling. |
| Fork-following height | For forks shorter than 1.2 m: approx. 350–400 mm. For forks longer than 1.2 m: approx. 400–800 mm, with Technical Support approval before installation. Measure from the fork top; confirm the exact datum in the installation plan. |
| Drop-down capture | Approx. 400 mm below the forks; fork-following height ranges do not apply. Raise to the pre-docking position. Detect only after full drop and the planned leg-to-optical-center height is reached. |
| Visibility | Check the full entry face, legs and beam at the nearest detection distance, the 1.2 m docking check point, and actual near, far and rack capture positions. Clear fork/fixture occlusion. Recheck images and valid clouds after height changes. |

Heights are empirical. Confirm forks around 1.2 m, special fixtures or conflicting height/visibility requirements with Technical Support. Recalibrate after moving the camera, replacing its bracket or changing its orientation.

## 2.4 Reference Point and Coordinates




Fork-entry reference point and forklift coordinate directions

1. The entry reference point lies on the lower edge of the upper beam at the entry face. Its lateral position is shown above; it is not the detection-box center or pallet volume center.
2. Forklift axes are shown in the diagram: +X rearward, +Y to the vehicle's right, and +Z upward. Position is in mm; angles are in °.
3. Do not mix camera and forklift coordinates. Agree on the reference point, axis directions and angle conventions before using results on the vehicle.
4. Verify directions with small fore/aft and lateral pallet movements, changing only one direction at a time.

**Zero values and validity** Negative values do not mean failure. Pitch is fixed at 0 as a placeholder, not a measured pitch angle.

# 3 Configuration and Calibration

## 3.1 Interface Overview

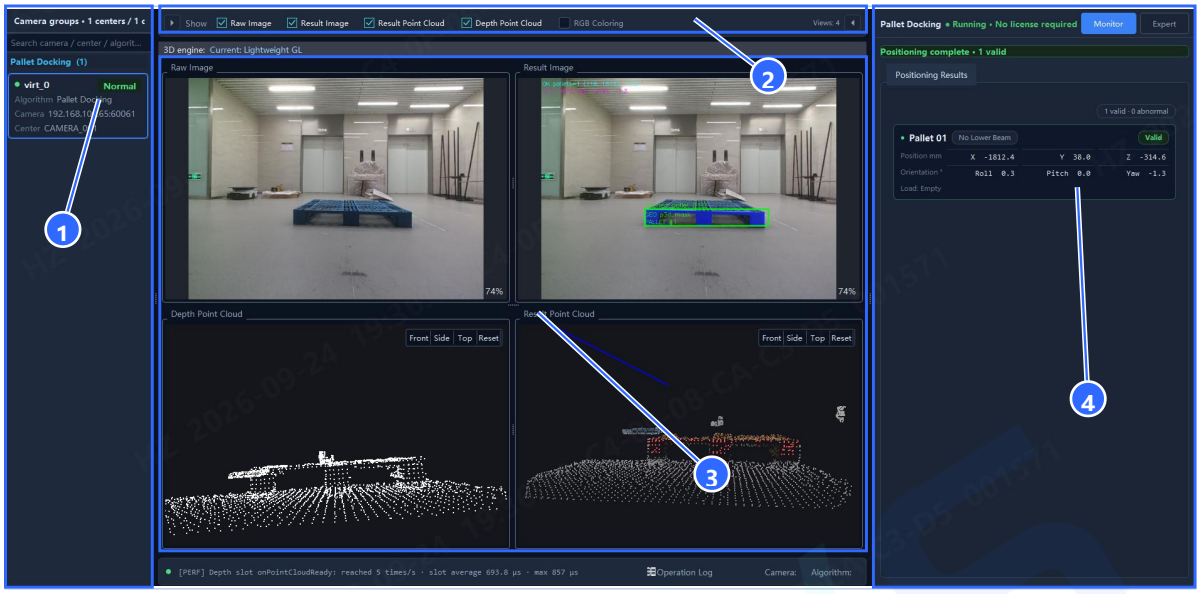



Figure 01 Main interface: ① Camera groups ② View controls ③ Images and point clouds ④ Algorithm panel

| Area | Purpose |
|---|---|
| ① Camera groups | Select [Pallet Docking]; check camera and center status. |
| ② View controls | Select the image and point-cloud views to display. |
| ③ Images and clouds | Inspect input and results. Front, Side, Top and Reset change the viewpoint only. |
| ④ Algorithm panel | [Monitor] shows results; [Expert] provides positioning, detection and log export. |

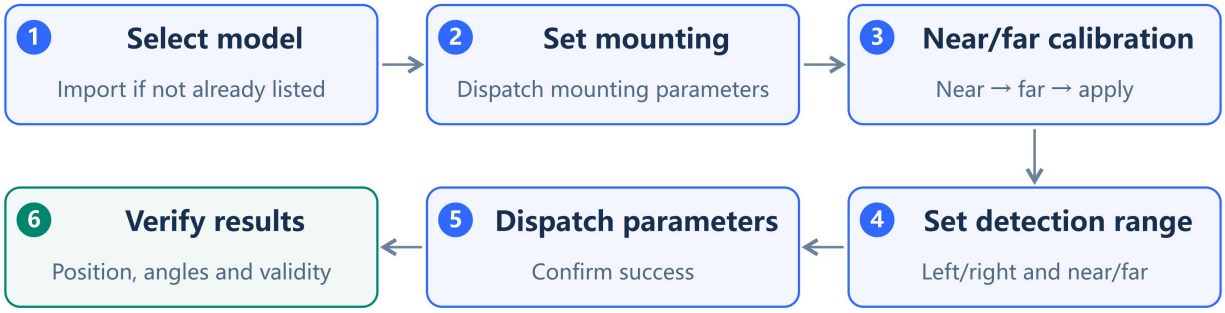
### Views and Point Clouds

| View | Purpose |
|---|---|
| Raw Image | Inspect targets, occlusion and lighting in the raw image. |
| Result Image | Check boxes or geometry against the actual target. |
| Depth Point Cloud | Check missing points, noise and target completeness. |
| Result Point Cloud | Compare fitted results with Depth Point Cloud. |
| RGB Coloring | RGB colors distinguish target/background; display only. |

Dispatch after changing settings; save templates before dispatch. For calibration, apply the result and wait for completion. Viewpoint, zoom and RGB coloring affect display only.

## 3.2 2D Semantic Positioning

Open [Expert] > [Positioning], switch to [2D], and confirm the current mode is [2D (Semantic)]. Follow the workflow below.




### 3.2.1 Import and Select a Model

Contact Technical Support for site-pallet RGB image requirements and the trained, validated project model. Import or select the model before installation settings and calibration.

1. In [Positioning], click [Select model...].

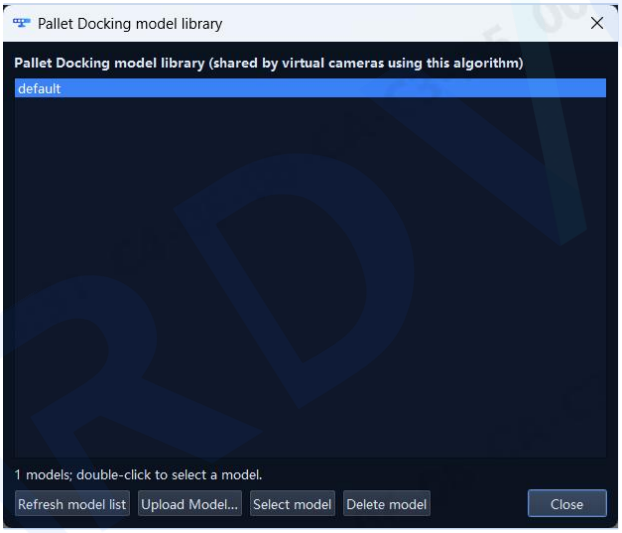



Figure 02 Pallet Docking model library

2. Upload if needed, refresh, then click [Select model] or double-click the model.

3. Check [Model Name] and click [Dispatch Parameters]. After success, continue with installation and calibration.

| Button | Function |
|---|---|
| Refresh Model List | Refresh the available models. |
| Upload Model... | Import a model file supplied for the project. |
| Select Model | Activate the selected model; double-click also selects it. |
| Delete Model | Delete only when no other task uses the model. |
| Close | Close; dispatch parameters on return. |

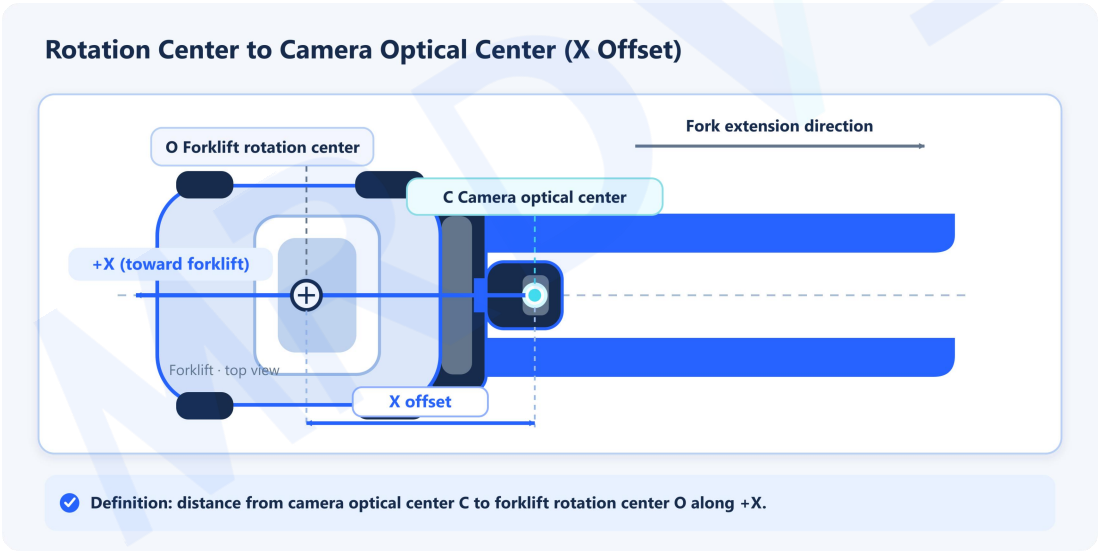
### 3.2.2 Installation and Calibration

#### Mounting Parameters

Open [Expert] > [Positioning] and set the following fields.

| Parameter | How to set |
| --- | --- |
| Camera orientation | Select the actual mounting orientation: 0°, 90°, 180° or 270°. |
| Rotation-to-optical-center distance (mm) | Enter the fore/aft distance between the rotation center and camera optical center. Positive: rotation center behind the camera; negative: in front. Enter 0 for direct camera use without this compensation. |
| Fork insertion depth (mm) | Adjust the final entry-reference position. Leave at 0 if no adjustment is needed. |
| Fork angle (°) | Adjust the final docking angle. Leave at 0 if no adjustment is needed. |

Click [Dispatch Parameters] and confirm success before calibration. Adjust insertion depth and fork angle during docking validation if needed, then dispatch again.




Measure the fore/aft distance between the rotation center and optical center

Ask the vehicle team to identify the rotation center; locate the optical center on the camera drawing. Recalibrate after changing orientation or this distance.

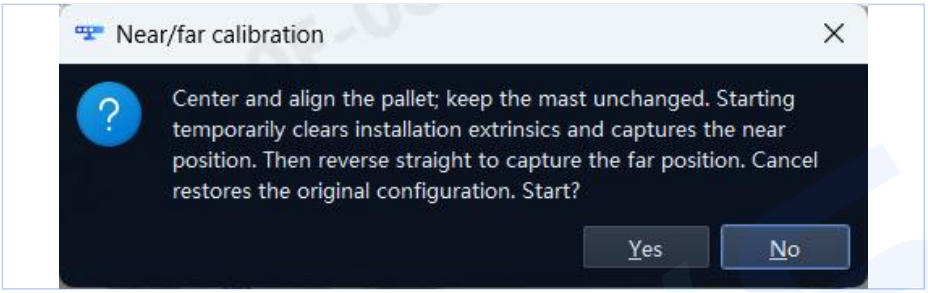
Near/far calibration corrects mounting angles and lateral offset. Set fore/aft mounting distance with the rotation-to-optical-center distance.

## Near/Far Calibration

Place one pallet on level ground and face it squarely. For near calibration, choose a position with the full entry face visible, stable recognition and clear image/depth data. The minimum recognition distance depends on pallet size and type. Keep the pallet, mast height/tilt and vehicle heading unchanged.

### 1 Calibrate Near

Stop the vehicle. Click [Calibrate near], then [Yes] in the confirmation dialog. Continue only after near capture completes and [Calibrate far] becomes available.




Near-calibration confirmation: click [Yes] to start or [No] to return.

### 2 Calibrate Far

Reverse straight 400–800 mm without turning or moving sideways. Stop and click [Calibrate far]. Wait for capture to finish and [Apply calibration] to become available. Click it and wait for the green completion message. The result is saved and dispatched automatically.

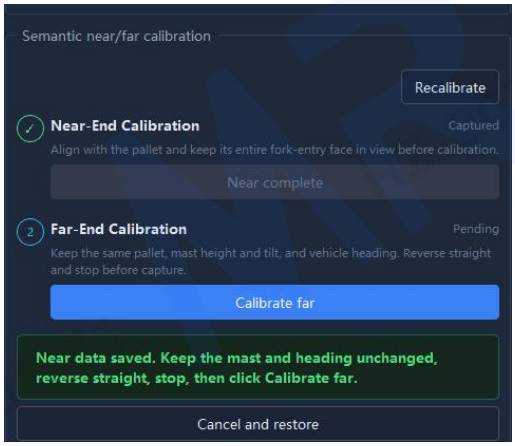



Figure 03 Near complete: ready for far capture

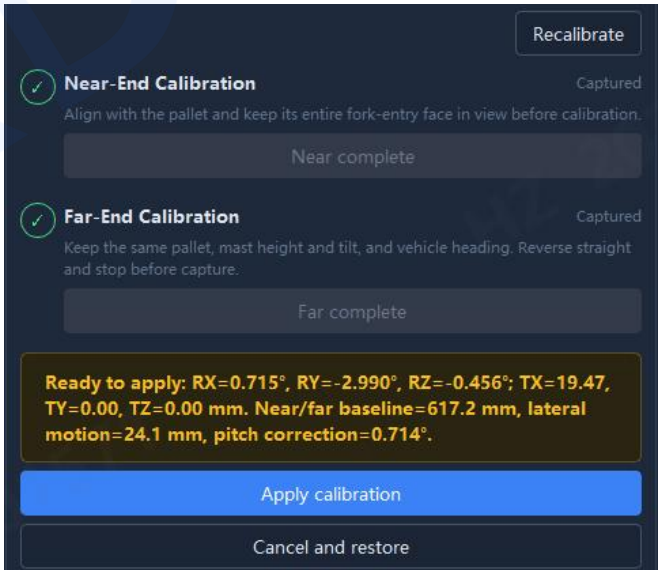



Figure 04 Far complete: ready to apply

If you exit or change the pallet, mast or vehicle heading, click [Cancel and restore] and start again from near capture.

### 3 Verify Results

Check that [Extrinsic Matrix] has updated. In [Monitor], check the same pallet at near, middle and far positions for position, angles and valid status.




Figure 05 Calibration complete and extrinsic matrix

Matrix fields: Pitch, Yaw and Roll correct mounting angles; Lateral is the left/right offset, Height the vertical offset, and Fore/aft the front/back offset. These are calibration parameters; Pitch in the positioning result remains fixed at 0.

If completion is not reported, do not proceed. Use Section 6.2 to troubleshoot the failed near or far capture.

### 3.2.3 Detection Range Settings

Set the left/right detection range and near/far distances, in mm.

| Setting | Meaning and setup |
| --- | --- |
| Lateral range (mm) | Set the left and right detection limits relative to the camera. |
| Detection distance (mm) | Set the nearest and farthest detection distances from the camera. |

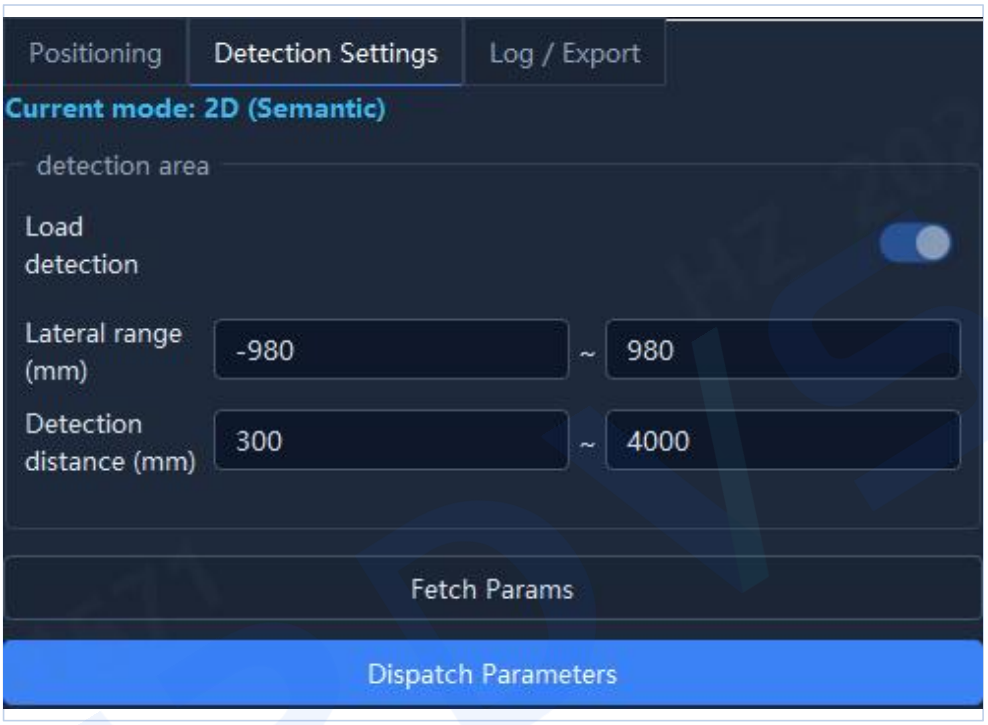



Figure 06 2D detection settings: lateral range, distance and parameter dispatch

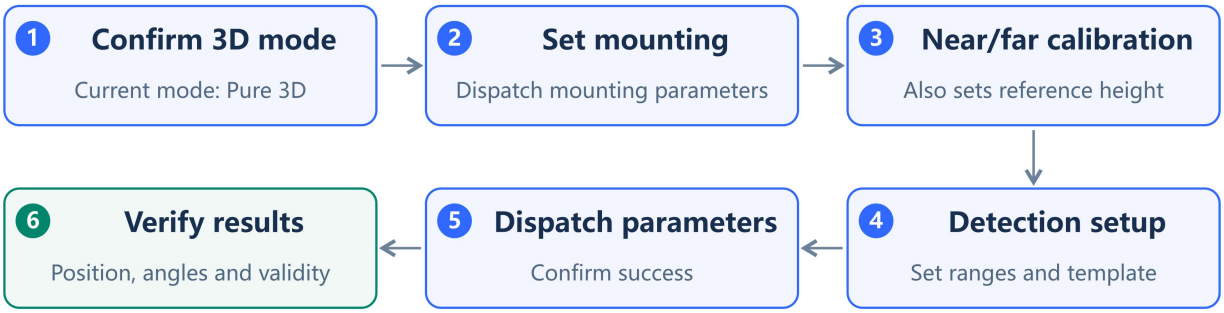
The Load detection control shown does not apply to 2D. Set only the lateral range and detection distance here.

1. Open [Expert] > [Detection Settings]. Click [Fetch Params] to view current settings.
2. Set lateral limits to cover the working width and exclude unwanted neighboring areas.
3. Set near and far limits, with the near value lower than the far value. 2D distance depends on the camera and site model; validate actual near and far positions.
4. Click [Dispatch Parameters] and confirm success. In [Monitor], check results inside, at and outside the limits.

**2D functions** 2D detects multiple eligible pallets in view without multi-pallet configuration. It does not support load detection or require a geometric pallet template.

## 3.3 Pure 3D Positioning

Open [Expert] > [Positioning], switch to [3D], and confirm the current mode is [3D (Pure 3D)]. Follow the workflow below.




### Capture height

Before calibration, set the camera to its actual height above the floor during detection. Near/far calibration also sets the reference height automatically.

> **Keep the 3D detection height fixed**
> During detection, keep the camera-to-floor height the same as during floor-reference calibration. **After changing detection height, auto-calibrate the floor reference again, dispatch parameters, and verify positioning before use.**

A capture-height change alone does not require near/far recalibration. Repeat full calibration if camera orientation or its mounting relationship to the vehicle changes.

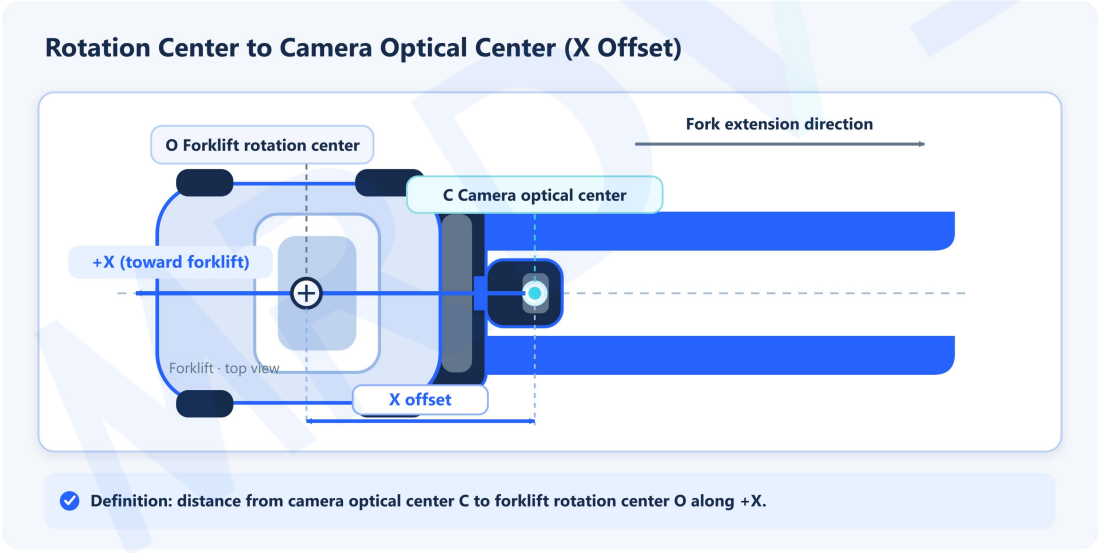
### 3.3.1 Installation and Calibration

#### Mounting Parameters

Open [Expert] > [Positioning] and set the following fields.

| Parameter | How to set |
| --- | --- |
| Camera orientation | Select the actual mounting orientation: 0°, 90°, 180° or 270°. |
| Rotation-to-optical-center distance (mm) | Enter the fore/aft distance between the rotation center and camera optical center. Positive: rotation center behind the camera; negative: in front. Enter 0 for direct camera use without this compensation. |
| Fork insertion depth (mm) | Adjust the final entry-reference position. Leave at 0 if no adjustment is needed. |
| Fork angle (°) | Adjust the final docking angle. Leave at 0 if no adjustment is needed. |

Click [Dispatch Parameters] and confirm success before calibration. Adjust insertion depth and fork angle during docking validation if needed, then dispatch again.




Measure the fore/aft distance between the rotation center and optical center

Ask the vehicle team to identify the rotation center; locate the optical center on the camera drawing. Recalibrate after changing orientation or this distance.

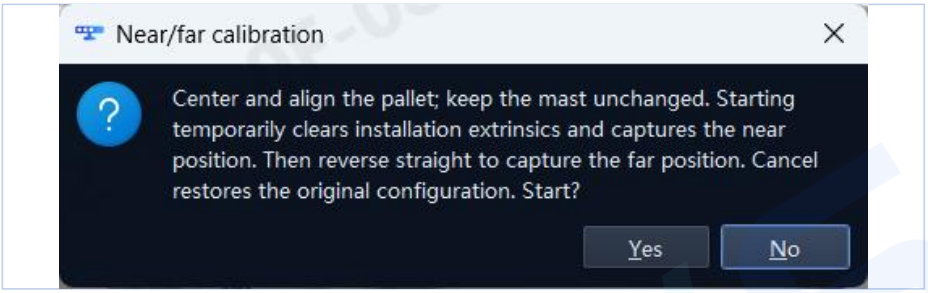
Near/far calibration corrects mounting angles and lateral offset. Set fore/aft mounting distance with the rotation-to-optical-center distance. Pure 3D also calibrates reference height.

## Near/Far Calibration

Use a three-leg pallet. Keep the floor and beam edges visible. Set the operating camera height and face the pallet squarely. For near calibration, choose a position with the full entry face visible and stable recognition. The minimum recognition distance depends on pallet size and type. Keep the pallet, mast height/tilt and vehicle heading unchanged.

### 1 Calibrate Near

Stop the vehicle. Click [Calibrate near], then [Yes] in the confirmation dialog. Continue only after near capture completes and [Calibrate far] becomes available.




Near-calibration confirmation: click [Yes] to start or [No] to return.

### 2 Calibrate Far

Reverse straight 400–800 mm without turning or moving sideways. Stop and click [Calibrate far]. Wait for capture to finish and [Apply calibration] to become available. Click it and wait for the green completion message. The result is saved and dispatched automatically.

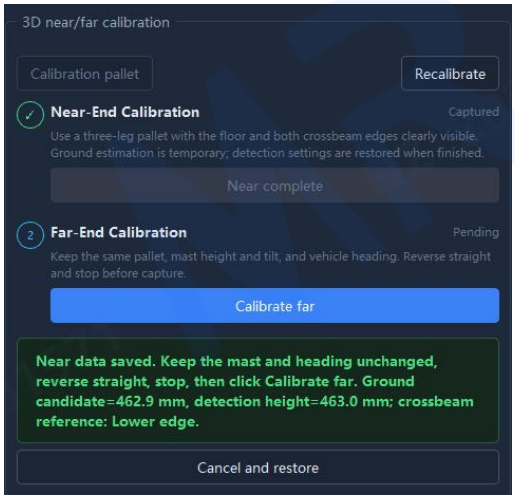



Figure 07 Near complete: ready for far capture

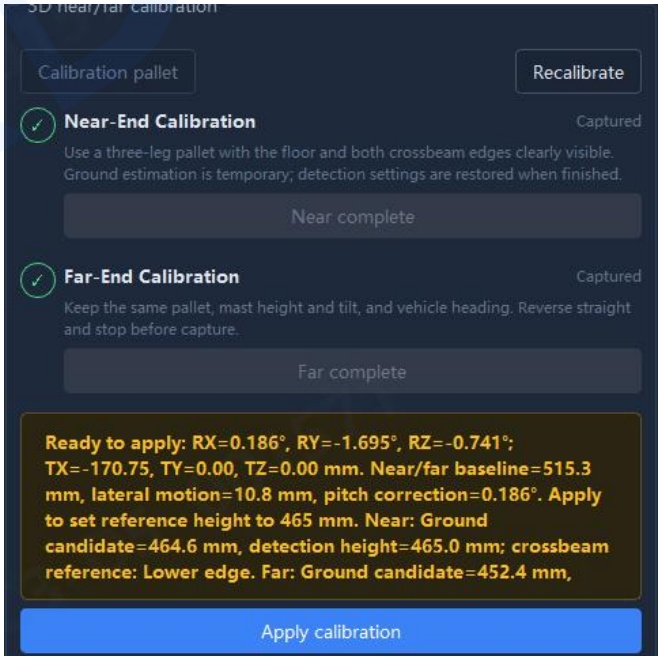



Figure 08 Far complete: ready to apply

If you exit or change the pallet, mast or vehicle heading, click [Cancel and restore] and start again from near capture.

### 3 Verify Results

Check that [Extrinsic Matrix] and [Reference Height] have updated. In [Monitor], verify positioning at near, middle and far distances. Use the calibrated height, not the screenshot value.




Figure 09 3D calibration complete and extrinsic matrix

Figures 08 and 09 are separate calibration examples; their values differ.

If completion is not reported, do not proceed. Use Section 6.2 to troubleshoot the failed near or far capture.

Matrix fields: Pitch, Yaw and Roll correct mounting angles; Lateral is the left/right offset, Height the vertical offset, and Fore/aft the front/back offset. These are calibration parameters; Pitch in the positioning result remains fixed at 0.

## Adjust the Calibration Pallet Only If Needed

If default settings do not recognize the pallet, first check full visibility and floor depth data, then compare measured dimensions. To adjust, click [Calibration pallet] in [Positioning]. Calibration requires a three-leg pallet.

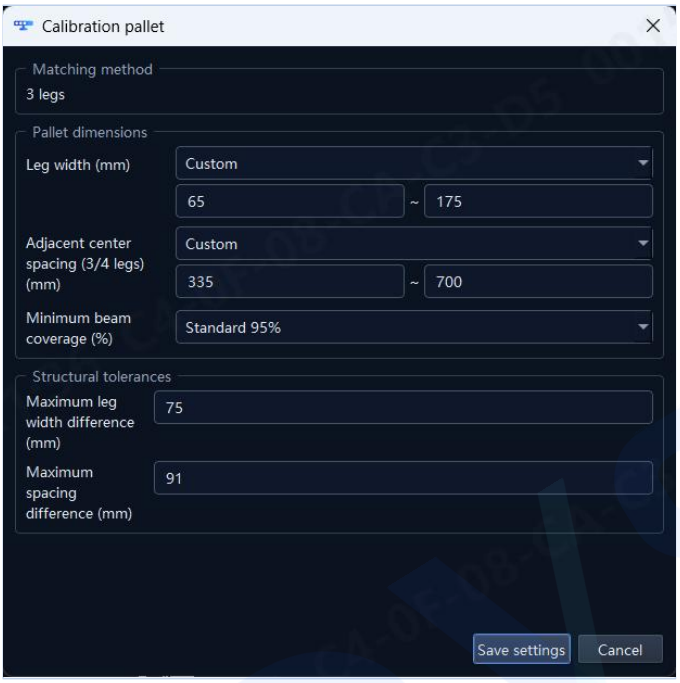



Figure 10 Calibration pallet settings

| Field | How to set |
|---|---|
| Leg width (mm) | Enter minimum and maximum visible width of one leg. |
| Adjacent center spacing (mm) | Set the allowed left–middle and middle–right center spacing, not the clear fork-opening width. |
| Minimum beam coverage (%) | Checks the beam between the legs. At 95%, at least 95% of horizontal sample positions must contain valid beam points. |
| Maximum leg width difference (mm) | Limits width differences between matched legs. |
| Maximum spacing difference (mm) | Limits center-spacing differences in the matched structure. |

Select a matching preset or enter measured ranges. Click [Save settings], return to the positioning page and click [Dispatch Parameters]. Restart near/far calibration from the near capture.

### 3.3.2 Detection Page and Detection Area

Complete calibration first, then configure detection settings in screen order. If capture height changes, recalibrate the floor using the steps on the next page.

1. Open [Detection Settings] and click [Fetch Params] to view current settings.

2. Select a matching [Current Template]. If none fits, add one as described in Section 3.3.3, then return here.

3. Configure the detection settings listed below.

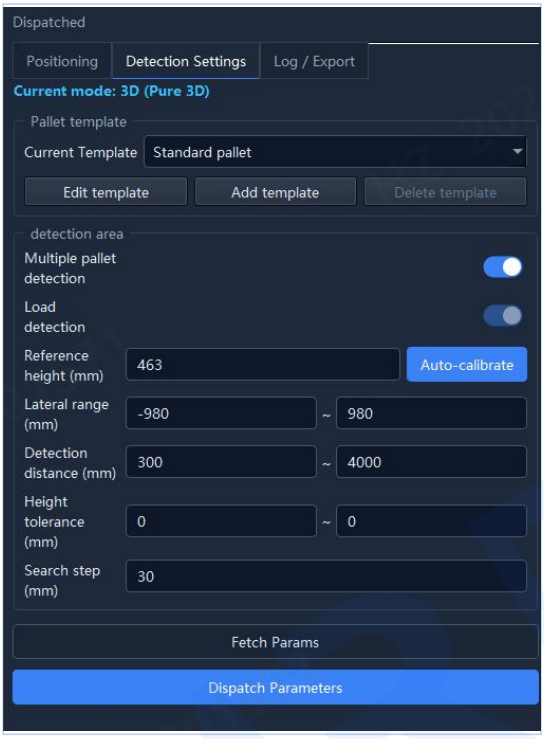



Figure 11 Pure 3D detection settings

| Setting | Meaning and operation |
|---|---|
| Multiple pallet detection | Detects stacked pallets. Check each layer for misses or duplicates. |
| Load detection | Indicates whether a pallet carries a load, not its weight, volume or stability. |
| Reference height (mm) | Detection reference height, obtained by calibration. |
| Lateral range (mm) | Set the left and right detection limits relative to the camera. |
| Detection distance (mm) | Set the nearest and farthest detection distances from the camera. |
| Height tolerance (mm) | Set how far to search above and below the reference height. |
| Search step (mm) | Vertical distance between successive search heights. |

## Save and Verify

1. Click [Dispatch Parameters] and confirm success.

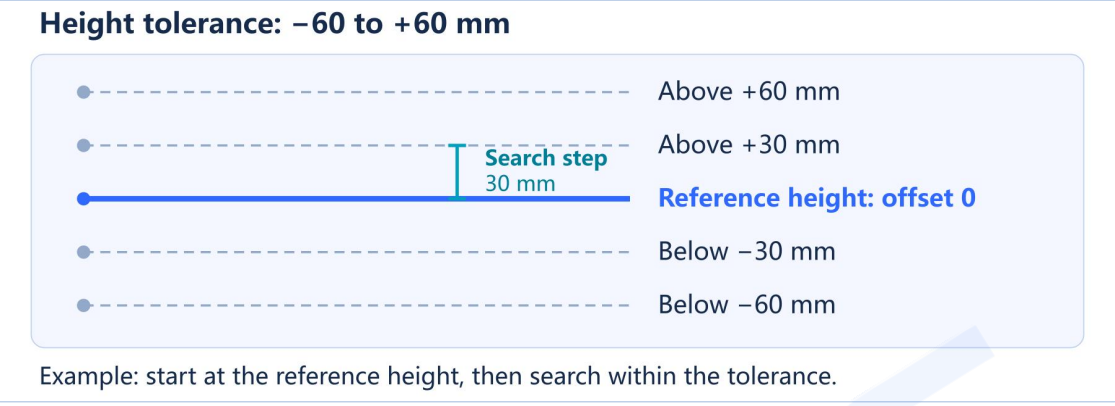
2. Open [Monitor] and test one pallet. Check that its status is [Valid] and its position and angles match the actual pallet.

3. For stacked pallets, enable [Multiple pallet detection] and dispatch again. Compare the results with the actual layers for misses or duplicates.

## Reference Height, Height Tolerance and Search Step

**Keep the 3D detection height fixed**

During detection, keep the camera-to-floor height the same as during floor-reference calibration. **After changing detection height, auto-calibrate the floor reference again, dispatch parameters, and verify positioning before use.**




| Parameter | Purpose and setting |
|---|---|
| Reference height | Floor reference for detection. Near/far calibration sets it; [Auto-calibrate] updates it separately. |
| Height tolerance | Search range below and above the reference. Both limits at 0 mean detection at the reference height only. |
| Search step | Interval between search heights. For example, 30 mm means one attempt every 30 mm. |
| **Detection mode** | **Search rule** |
| Single pallet | Try the reference height first, then step above and below within the tolerance. Stop when a pallet is found. |
| Multiple pallets | Does not use the height tolerance or search step on this page. |

## Recalibrate the Floor after Changing Capture Height

1. Fix the camera at its actual detection height, stop the vehicle and check that the floor point cloud is valid.

2. In [Detection Settings], click [Auto-calibrate] beside [Reference Height] and wait for the value to update.




3. Click [Dispatch Parameters]. After success, verify pallet positioning in [Monitor].

If camera orientation or its mounting relationship to the vehicle changes, repeat near/far calibration.

### 3.3.3 Detection Templates and Parameters

## Add, Edit and Delete Templates

1. Open [Expert] > [Detection Settings]. Click [Add Template] and enter [Template Name]. To edit an existing template, select it first and click [Edit Template].

Configure one common template to cover the pallets in the project. Use matching presets for leg width, center spacing and beam coverage, or [Custom] ranges based on measurements. Only when pallets differ too much to share a template should you create another and switch manually. Parameters follow screen order.




Figure 12 Edit a detection template

2. Select the actual leg count and set dimensions and structural tolerances. The following pages explain the parameters.

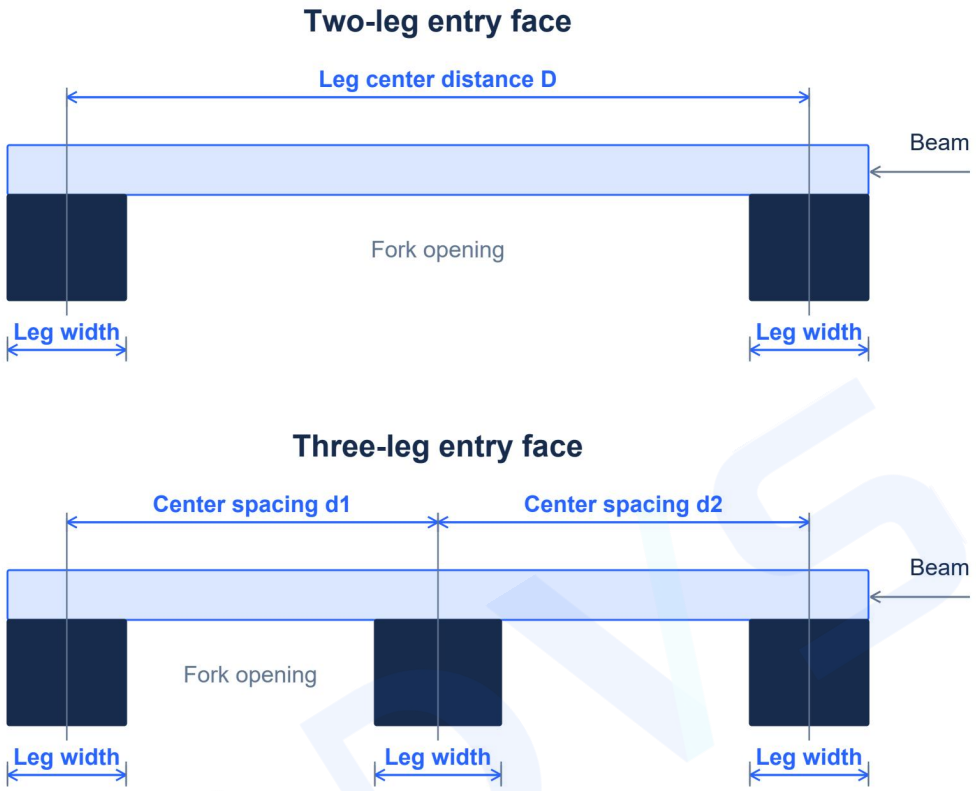
3. For a new template, click [Save template]; for an existing one, click [Save parameters].

4. Return to the detection page. Check [Current Template], click [Dispatch Parameters], and verify with actual pallets.

**To delete a custom template, select it in [Current Template], click [Delete Template], check its name and confirm. Then select the template to use and dispatch parameters. Built-in templates cannot be deleted.**

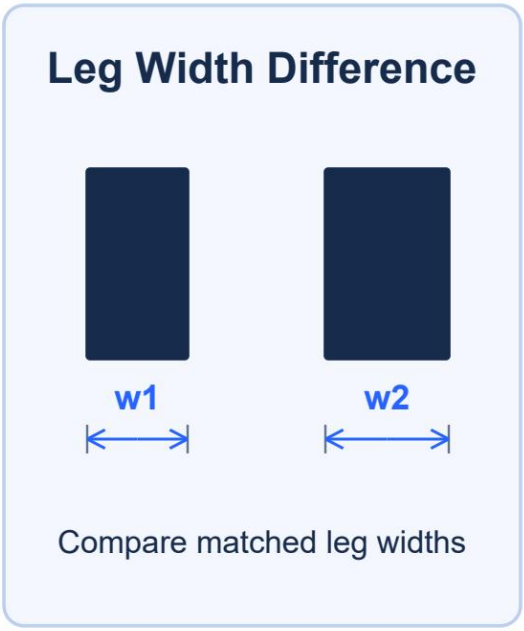
## Dimensions

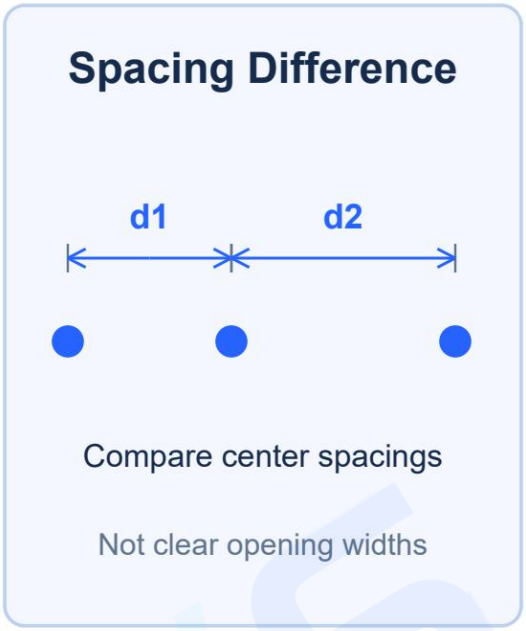
Open [Add Template] or [Edit Template] in [Detection Settings] and complete the fields below. Measure spacing between leg centers, not the clear fork opening.




| Parameter | How to set |
| --- | --- |
| Leg count matching | Select 2, 3 or 4 legs to match the fork-entry face. Multiple selections are allowed; select at least one. |
| Leg width (mm) | Enter the minimum and maximum visible width of one leg, not the full pallet width. |
| Adjacent center spacing (3/4 legs) | Set the range between adjacent leg centers. Four-leg matching uses the upper limit. |
| Leg center distance (2 legs) (mm) | Set the distance range between the two leg centers. Default: 700–1600 mm. |
| Minimum beam coverage (%) | Checks the beam between the legs. At 95%, valid beam points must be found at least 95% of the horizontal sample positions. Lower it for thin or interrupted beams, then check for false pallet detections in the background. |

### Structural Tolerances




| Parameter | How to set |
| --- | --- |
| Maximum leg width difference (mm) | Set the largest permitted width difference between matched legs. |
| Maximum spacing difference (mm) | Set the largest permitted difference between matched center spacings. |

Enter structural tolerances as numeric values. Change one parameter group at a time, save and dispatch, then check for missed pallets and false background detections.

## 3.4 Monitor and Verify Positioning Results

After dispatch succeeds, open [Monitor]. Check that images keep updating and detection boxes identify the intended pallets. Then read the result cards.

### 2D Semantic Monitor Page

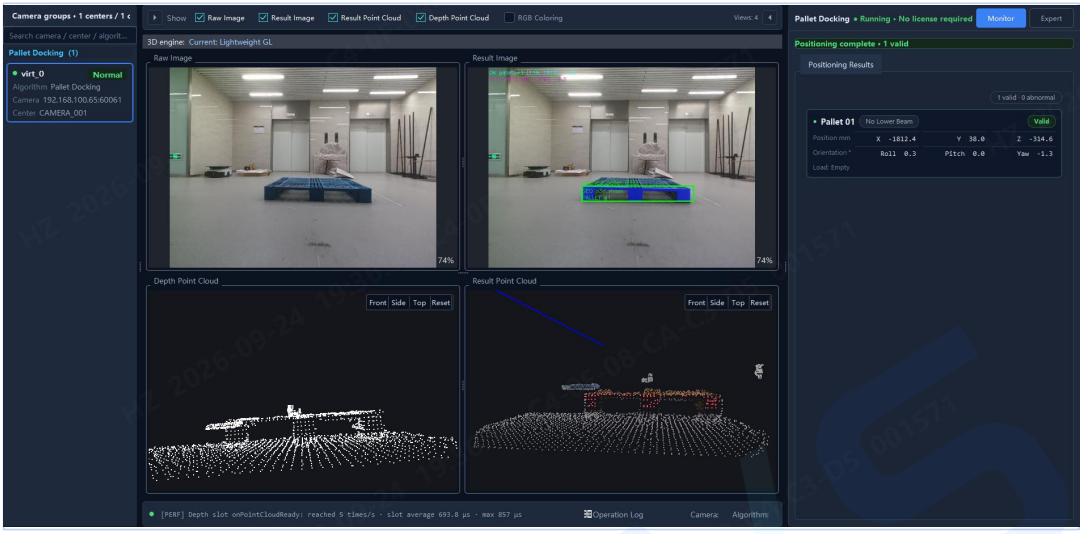



Figure 13 2D Monitor: semantic results, point clouds and per-pallet result cards

### Pure 3D Positioning Monitor

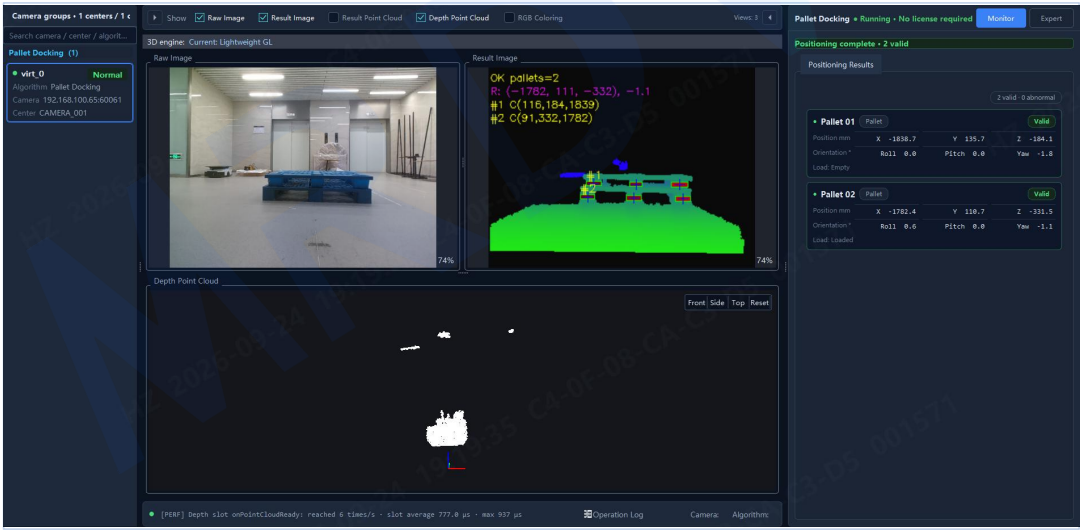


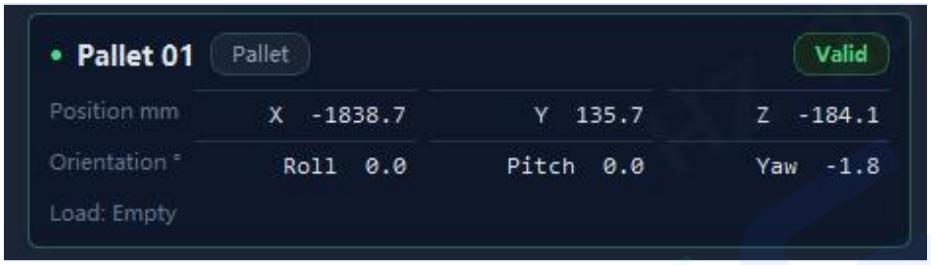

Figure 14 3D monitor: point-cloud matching and per-pallet result cards

### 3.4.1 Read Status and Result Cards

2D result card: class, validity, position and orientation




3D result card: validity, position and orientation; load status when enabled




| Item | Description |
|---|---|
| Valid / abnormal count | Counts current positioning results. Inspect individual cards when errors occur. |
| Green valid status | The pallet has a valid positioning result. Still confirm it is the correct target. |
| Yellow abnormal status | Read the error message. Do not use a pose whose position or orientation is shown as “—”. |
| Position X / Y / Z | Entry reference position in forklift coordinates, in mm. |
| Roll / Pitch / Yaw | Valid orientation outputs are Roll and Yaw, in °. Pitch is fixed at 0 and is not a valid output. |
| Load | Use only in 3D with load detection enabled. In 2D, this field does not indicate whether the pallet carries a load. |

**Status distinction** “Positioning complete” means processing has ended, not that every target was positioned successfully.

Card numbers identify the current results. They are not persistent IDs or picking priorities.

### 3.4.2 Invalid Positioning Results

Do not use these results for docking. Check positioning status even when a detection box is present.

| Result | Meaning | Action |
| --- | --- | --- |
| Positioning Error | No valid pose. The example shows an entry-face angle over 15° relative to the camera. | Realign the pallet or camera to reduce the angle, then detect again. |
| Incomplete Structure | In 2D, occlusion prevents recognition of the complete entry face. | Remove the obstruction, show the full entry face and detect again. |



Figure 15 Detection box with pallet rotation

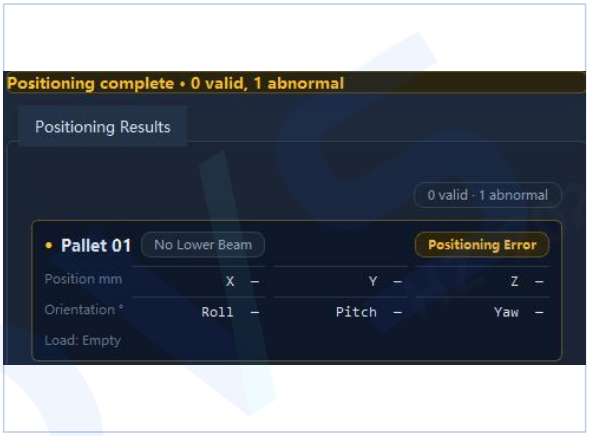



Figure 16 Positioning Error: pose unavailable

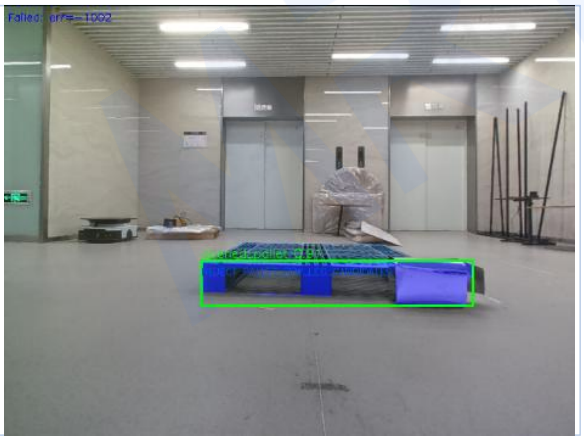


Figure 17 Occluded entry-face structure

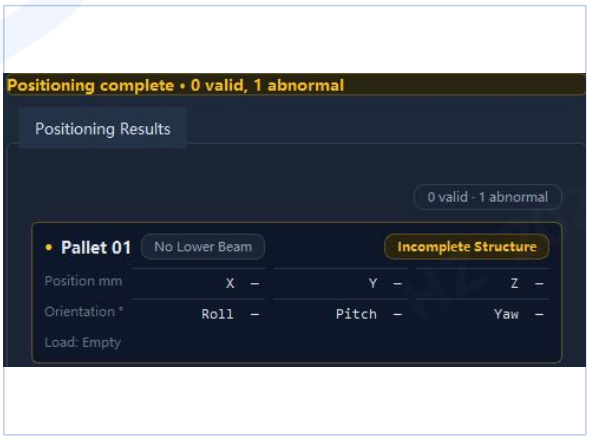
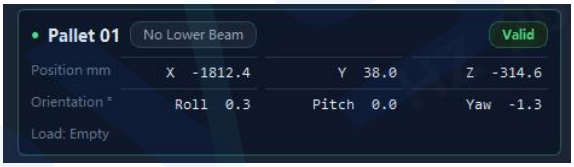



Figure 18 Incomplete Structure: pose unavailable

### 3.4.3 2D Pallet Structure Classification

In 2D mode, the model detects whether the fork-entry face has a lower crossbeam. Classification is automatic with a suitable model; no separate switch is required.

In [Monitor], compare the class label on the pallet result card with the actual fork-entry face.

| **With Lower Beam**<br>**Closed pallet** | **No Lower Beam**<br>**Open pallet** |
|---|---|
|  |  |
| A continuous crossbeam joins the legs below the openings. | No continuous crossbeam below the fork openings. |
|  |  |

Pallet structure and matching result labels

Here, Closed pallet corresponds to [With Lower Beam], and Open pallet to [No Lower Beam]. The lower beam is the bottom board spanning beneath the fork openings.

Classification describes only the entry face seen by the camera, not the whole pallet base. Different entry faces of the same pallet may differ. Only 2D provides this classification; check positioning status separately.

2D mode does not support load detection. Ignore the [Load] field in these screenshots.

### 3.4.4 Verify Positioning and Docking

| Test scenario | What to check |
| --- | --- |
| Near, middle and far | Stable results for the same pallet, without clear distance-related drift in lateral position or angle. |
| Lateral and fore/aft motion | Result direction and magnitude match the coordinate convention. |
| Different angles | Reported orientation matches the pallet and meets project tolerances. |
| Stacked pallets | No missing or duplicate layers; the vehicle selects the correct target layer. |
| No target and occlusion | The vehicle stops using old results when the target is removed or the pose is invalid. |
| Load status | Test empty and loaded pallets only in 3D with load detection enabled. Do not use this judgment in 2D. |
| External communication | Target, status and pose agree between sender and receiver. |

#### If Deviations Occur

1. Check input views, target correspondence, mode and model or template.
2. Check mounting orientation and calibration conditions; recalibrate if needed.
3. Adjust insertion depth and angle only after mounting geometry is correct. Do not use offsets to compensate for invalid calibration.

Record versions, model or template, calibration settings and results. Verify accuracy and repeatability against project criteria. Use Section 2.2 to set vehicle wait times and timeouts with margin. Validate result updates for the mode in use; validate both modes if both are deployed.

## 3.5 Daily Operation

1. Check the lens, bracket and cables before work; confirm the mount is secure.
2. Start AW3, select the correct camera group, and check continuous streaming and normal algorithm operation.
3. Use a known pallet to check position, direction and validity.
4. Confirm vehicle-side result reception and safety interlocks before automatic operation.

**Stop conditions** Stop using a target pose if the camera is offline, frames freeze, positioning fails, the target is wrong or results expire. Recheck after recovery.

# 4 Using the Functions

This chapter covers pallet template management, multiple-pallet detection and load detection.

## 4.1 Manage Pallet Templates

Use one common template for the project wherever possible. Create another and switch manually only when pallet differences prevent sharing. Templates can be added, edited and deleted; see Section 3.3.3.

## 4.2 Multiple Pallet Detection

Use this function when multiple pallets are present in one view. 2D needs no multi-pallet switch. In 3D, enable [Multiple pallet detection] to search stacked pallets layer by layer.

1. 2D: select a suitable model and inspect each pallet's result. No leg-count or other geometric setup is needed.
2. 3D: open [Expert] > [Detection Settings], enable [Multiple pallet detection], click [Dispatch Parameters].
3. In [Monitor], compare actual pallet count and layers. Check misses, duplicates and target correspondence.

Missing layers: check ranges, occlusion and point clouds. Duplicates: check overly broad templates and false structure matches. Match each result to a physical layer. The vehicle selects the target and motion; do not use list order or card numbers.

## 4.3 Load Detection

Pure 3D only: [Load detection] checks whether the pallet carries a load.

1. Confirm stable pure 3D positioning and a template that matches the pallet.
2. Open [Expert] > [Detection Settings], enable [Load detection], and click [Dispatch Parameters].
3. Test empty and loaded pallets separately; check load status on the [Monitor] result card.
4. Revalidate after changing goods or occlusion. If results disagree with the scene, retain images, clouds and logs.

| Condition | Note |
| --- | --- |
| 2D mode | Load detection is not supported. Do not use the displayed load field to judge empty or loaded. |
| Disabled or unknown status | Do not treat this as empty. |
| Positioning error | Restore valid pallet positioning before using load status. |
| Use of result | Indicates load presence, not weight, volume or load stability. |

# 5 Communication and Result Data

## 5.1 Configure and Verify Communication

AW3 manages communication. Follow Section 7.3 of the AW3 Deployment User Manual to configure trigger listening and result forwarding.

Trigger one detection from the receiver and check the received pallet count, positioning status, position and angles.

## 5.2 Read Positioning Results

Check overall validity and the pallet status before reading position and angles. pallets[] refers to each returned pallet.

| Result | Field | Meaning |
|---|---|---|
| Overall validity | valid | true means the result is valid. |
| Pallet count | count | Number of returned pallets. |
| Positioning status | pallets[].ret_code | 0 means this pallet was positioned successfully. |
| Error description | pallets[].error_str | Failure reason; empty on success. |
| X position | pallets[].forklift_pose.x | mm; positive points backward. |
| Y position | pallets[].forklift_pose.y | mm; positive points to the vehicle’ s right. |
| Z position | pallets[].forklift_pose.z | mm; positive points upward. |
| Roll angle | pallets[].forklift_pose.roll | Roll in degrees. |
| Yaw angle | pallets[].forklift_pose.yaw | Yaw in degrees. |

Pitch is fixed at 0 and is not a valid measured output.

## 5.3 Result JSON Example

Main fields from a successful 2D positioning result.


```
{
  "valid": true,
  "ret_code": 0,
  "error_str": "",
  "count": 1,
  "pallets": [
    {
      "pallet_type": "opened_pallet",
      "ret_code": 0,
      "error_str": "",
      "forklift_pose": {
        "x": -1304.7950439453125,
        "y": -9.663567543029785,
        "z": -320.1379699707031,
        "yaw": 1.6952972412109375,
        "pitch": 0.0,
        "roll": 0.4580188989639282
      }
    }
  ]
}
```


The example is successful and returns one Open pallet with no lower beam (opened_pallet). Forklift coordinates: X = −1304.795 mm, Y = −9.664 mm, Z = −320.138 mm; Roll = 0.458°, Yaw = 1.695° (rounded to three decimals here).

| | |
|---|---|
| **Class • 2D only** | pallets[].pallet_type: opened_pallet means No Lower Beam; closed_pallet means With Lower Beam. Do not use this field for lower-beam classification in 3D. |
| **Load • 3D only** | pallets[].pallet.is_onload: with load detection enabled and valid positioning, true means loaded and false means empty. Disabled, missing or null is not empty. Do not use this field in 2D. |

# 6 Troubleshooting and Maintenance

First check that Raw Image and Depth Point Cloud keep updating. Restore streaming if input fails; otherwise investigate positioning performance.

## 6.1 System and Streaming Errors

| Symptom | Checks and action |
| --- | --- |
| No image or frozen frames | Check camera power, cable, network and central-service status. Confirm the selected camera group. |
| Camera online, no valid depth | Check distance, reflections, lens contamination and cloud quality. Adjust capture conditions per the hardware manual. |
| Algorithm stopped or license error | Check algorithm status, version and license messages. Follow the AW3 Deployment User Manual. |
| Streaming fails after camera setup | If LxCameraViewer was just used, open [Device Connection / Base Camera] > [Central Service] > [Restart current center]. |
| AW3 has results, receiver does not | Check protocol, address, port, receiving service and camera group. Verify as in Section 5.1. |

### After Recovery

1. Confirm Raw Image and Depth Point Cloud update continuously rather than show an old frame.
2. Check positioning with a known pallet.
3. Confirm the receiver gets new results after recovery before resuming work.

**Operating rule** Do not continue fork insertion with an old pose after disconnection, frozen frames or an algorithm error.

## 6.2 Positioning Performance Issues

See Section 3.4.2 for invalid-result meanings and examples. Use this table to find the corrective action.

| Message or issue | Action |
|---|---|
| Positioning Error | Align the entry face so its rotation relative to the camera does not exceed 15°, then detect again. |
| Incomplete Structure | Remove occlusion, make the full entry face visible, and detect again. |
| No pallet recognized in 2D | Confirm model support; check the image, depth and detection range. |
| No pallet recognized in 3D | Check leg count, dimensions, beam coverage, reference height and range. Reposition visibly tilted pallets. |
| Reflective or black pallet | Reflective surfaces are unsuitable. Assess black or low-reflectance pallets on site; check point-cloud completeness, not RGB appearance alone. |
| Background matched in 3D | Narrow the detection area and tighten template dimensions and tolerances. |
| Missing or duplicate pallets | Check occlusion and clouds. In 3D, also check multi-pallet detection and template. |
| Near calibration failed | Stop and face the pallet squarely. Check the 2D model, or the 3D three-leg settings and floor cloud. |
| Far calibration failed | Complete near capture first. Reverse straight 400–800 mm, stop and capture; keep mast position and heading fixed. |
| Unreasonable calibration or position drift | Check mounting orientation and rotation-center distance. Cancel and restore, realign and recalibrate. |
| Incorrect load state | Confirm load detection is enabled and positioning is valid. Clear occlusion, check clouds and retain misclassified samples. |
| Parameters not applied | Confirm dispatch succeeded. Fetch parameters and check camera group, mode and template. |

**Troubleshooting method** Change one parameter group at a time. For recurring issues, save settings and corresponding data, then export as in Section 6.3.

## 6.3 Export Troubleshooting Data

1. Open [Expert] > [Log / Export]. Record fault time and the full message; click [Copy All] if needed.

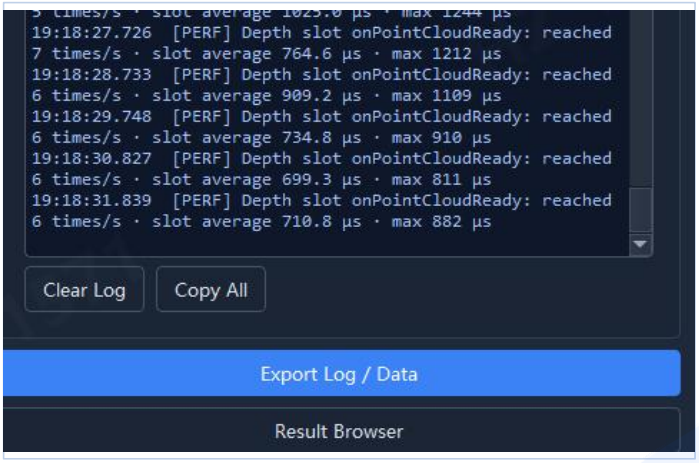



Figure 19 Log and export controls

2. Click [Export Log / Data]. Follow Chapter 8 of the AW3 Deployment User Manual to select algorithm, category and a time range covering before and after the fault.

3. Use [Result Browser] to check result images and JSON. Confirm export completeness before sending to Technical Support.

Do not clear logs before export. Include versions, configuration, site photos, pallet dimensions and distance, failure time and reproduction steps. Verify the exported images, clouds and logs.

## 6.4 Maintenance and Revalidation

| When | Requirement |
|---|---|
| Daily maintenance | Check lens, bracket, fasteners, cables, power and cooling. Clean per the hardware manual without moving the mounted camera. |
| Camera or mounting changes | Recheck orientation and camera-to-fork geometry. Complete near/far calibration. |
| 3D capture height changes | Recalibrate the floor reference height with [Auto-calibrate], click [Dispatch Parameters], then verify pallet positioning. |
| Model or software update | Back up settings, check versions and the model report, then test representative on-site pallets. |
| Different pallet type | 2D: check model coverage. 3D: configure a matching template. |
| Scene or mast conditions change | Check calibration still applies. Verify depth quality, ranges, position and direction. |

**Keep records** Retain validated models, templates and settings for post-maintenance comparison and recovery.

# Revision History and Technical Support

| Version | Date | Revision |
| --- | --- | --- |
| V0.1 | 2026-09-28 | V0.1: consolidated repeated workflows, monitoring and parameter guidance; retained full functionality and troubleshooting; aligned 2/3/4-leg rules and the 400–800 mm calibration reverse distance. |

## Technical Support

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Scan to open MRDVS on GitHub